

Reflection Paper

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DAMO 699-2: Capstone Project

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My Capstone Project is titled *A Unified Framework for Multi-Tenant Cluster Management with Overcommit and Guaranteed Service Level Agreement*. It is a two-layer system that decides how to pack many tenants and their jobs onto a shared pool of machines while keeping memory use safe and meeting service level agreements. A prediction layer forecasts demand, and the optimization layer that I built turns those forecasts into real-time and plan-ahead scheduling decisions.

Initial Expectations and Goals

When I started the Capstone Project, I expected the optimization work to be easy. I thought a single, simple optimization model would solve the whole problem on its own. I pictured one clean formulation that I could write, run, and finish. I chose this Capstone because I want to work in the optimization analytics field after I finish school, and I felt this project would give me the hands-on experience I needed to get there.

My personal goals were focused on skill building. I wanted to learn how to write optimization models for a complex system rather than for a textbook example. I wanted to understand how scientists and professionals actually develop optimization solutions for real world problems. Most of all, I wanted to learn the full process, which is how to reason about a problem, turn it into a model, build the model in code, and then test it.

Looking back, my first expectation was the most important one to correct. The belief that one simple model would be enough set the tone for how I planned my early work. When that belief broke down, I had to learn to plan differently and to accept that a real solution would take several stages. That shift in mindset turned out to be one of the most valuable outcomes of the whole project.

Role and Contributions

The Capstone Project had two layers. There was a prediction layer and an optimization layer. I chose to work on the optimization layer while the rest of the team worked on the prediction layer. My job was to create the mathematical models, write the code, and test the code that turned predicted demand into real scheduling decisions.

My main responsibility was the Real-Time Optimization Model, which acts as the scheduler. I worked on the math for the model, developed the code, and tested it. I also took responsibility for refactoring the model when the analytical results showed it could be improved. Later I took on a second model, the Plan-Ahead Model, which we added to give the project more depth. I followed the same process for that model, starting from the math and ending with tested code. On top of the two models, I built a visual simulation that shows the entire pipeline running end to end, so that anyone could see how the prediction layer and the optimization layer fit together.

Challenges and Problem-Solving

I faced challenges in communication and in technical limitations. Time management was not a major problem for me. The communication challenge was that my teammates sometimes did not understand the technical context I was explaining. The two layers used different ideas, and the optimization side carried a lot of math that was not obvious to everyone.

The technical limitation appeared later, when I started doing deep computational time analysis on the models. I found that both models struggled when the parameters grew large. The solver took too long, and in some cases it ran out of memory before it finished.

I addressed the communication challenge by sharing documents that broke down the complex topics into smaller pieces. When that was not enough, I met with my teammates and walked them through the ideas visually until they understood. I addressed the technical limitation

by developing a decomposition heuristic in the code. Instead of solving one large problem, the heuristic breaks the problem into smaller pieces and solves them in sequence. This let the models run in a satisfactory time even when the parameters were large.

Solving the limitation taught me more than solving the original model did. I had to study why the solver slowed down, test the models at different sizes, and measure where the cost grew the fastest. Only after I understood the cause could I design a fix that kept the results trustworthy while making the run time practical. That cycle of measure, diagnose, and redesign is a skill I will carry into any future optimization work.

Learning and Development

I gained several new skills and tools during this project. I learned how to build a simulation dashboard using a backend and an API. I learned how to use commercial solvers such as Gurobi. I also gained an important insight, which is that real world problems are not solved by basic model math once the scale gets large. At scale, you need heuristics to get a good answer in a reasonable amount of time.

The project greatly improved my understanding of the field. Multi-tenant memory utilization is a real world problem that businesses want to solve, because it controls cost and service quality at the same time. Working on it showed me how optimization analytics connects directly to business decision-making, and how a good model can turn a vague operational goal into a concrete and testable plan.

Teamwork and Communication

I would evaluate our team dynamics as steady and reliable. My teammates were responsible for managing their part, which helped keep the project on pace. We communicated

when we needed to, and we kept each other informed at the points where the two layers had to meet.

I learned that teamwork can be challenging when you share ideas that not everyone understands. Having more people gets the work done faster, but it also creates more friction that the team has to work through. The lesson for me was that clear communication is just as important as technical skill, because a good model has no value if the rest of the team cannot follow how it works.

Self-Assessment and Future Applications

I grew professionally during this project. I learned to take things one step at a time and to overcome problems incrementally instead of trying to solve everything at once. I started with one simple model, and I ended with two advanced models that use heuristics, plus a simulation dashboard that ties the whole pipeline together. That progression is the clearest sign of my growth.

This experience will shape my future studies and my career. It has given me the confidence to put optimization expert on my resume as I look for a job after school. More than the title, it changed how I approach problems. I now expect real problems to be harder than they first appear, and I know how to break them down, build a solution, and test it until it works.